

## Solutions #W7

Problem 1: let  $y = \sum_{m=0}^{\infty} a_m x^m$

$$y' = \sum_{m=1}^{\infty} a_m m x^{m-1}, \quad y'' = \sum_{m=2}^{\infty} a_m m(m-1) x^{m-2}$$

Substituting the series in the ODE, we get

$$2a_2 + 6a_3x - (a_1 + 2a_2x) + \sum_{m=4}^{\infty} ((m-1)ma_m - (m-1)a_{m-1} + a_{m-4})x^{m-2} = 0$$

$$\Rightarrow 2a_2 - a_1 = 0 \Rightarrow a_2 = \frac{a_1}{2}$$

$$6a_3 - 2a_2 = 0 \Rightarrow a_3 = \frac{a_2}{3} = \frac{a_1}{3!}$$

when  $m=4$

$$12a_4 - 3a_3 + a_0 = 0 \Rightarrow a_4 = \frac{3a_3 - a_0}{12} = \frac{a_1}{4!} - \frac{a_0}{12}$$

when  $m=5$

$$20a_5 - 4a_4 + a_1 = 0 \Rightarrow a_5 = -\frac{a_0}{2} - \frac{a_1}{5}$$

The general solution is then

$$y = a_0 \left( 1 - \frac{1}{12} x^4 - \frac{1}{60} x^5 \dots \right) + a_1 \left( x + \frac{1}{2!} x^2 + \frac{1}{3!} x^3 + \frac{1}{4!} x^4 - \frac{1}{5!} x^5 \dots \right)$$

Problem 2: let  $u = \frac{x}{a}$

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = \frac{dy}{du} \frac{1}{a}$$

$$\frac{d^2y}{dx^2} = \frac{d}{du} \left( \frac{dy}{du} \frac{1}{a} \right) \frac{du}{dx} = \frac{d^2y}{du^2} \frac{1}{a^2}$$

Substituting into the differential equation, we get

$$(a^2 - a^2 u^2) \frac{d^2y}{du^2} \frac{1}{a^2} - 2(au) \frac{dy}{du} \frac{1}{a} + n(n+1)y = 0$$

$$(1 - u^2) \frac{d^2y}{du^2} - 2u \frac{dy}{du} + n(n+1)y = 0$$

whose general solution is

$$y(u) = c_1 y_1(x) + c_2 y_2(x)$$

$$y(x) = c_1 y_1\left(\frac{x}{a}\right) + c_2 y_2\left(\frac{x}{a}\right)$$

Problem 3 :

If  $z = \sqrt{x}$ , then

$$z' = \frac{1}{2\sqrt{x}} = \frac{1}{2} x^{-1/2}$$

$$y' = \frac{dy}{dz} z' = \frac{dy}{dz} \frac{1}{2} x^{-1/2}$$

$$\begin{aligned} y'' &= \frac{dy'}{dz} \frac{dz}{dx} = \frac{1}{2} \left( -x^{-3/2} \frac{dy}{dz} + x^{-1/2} \frac{d^2y}{dz^2} \right) \frac{1}{2} x^{-1/2} \\ &= \frac{1}{4} \left( x^{-2} \frac{d^2y}{dz^2} - x^{-3} \frac{dy}{dz} \right) \end{aligned}$$

With these, we can rewrite the ODE as

$$\frac{z^2}{4} \left( z^{-2} \frac{d^2y}{dz^2} - z^{-3} \frac{dy}{dz} \right) + \frac{1}{2} z^{-1} \frac{dy}{dz} + \frac{1}{4} y = 0$$

$$\frac{1}{4} \frac{d^2 y}{dz^2} - \frac{1}{4} z^{-1} \frac{dy}{dz} + \frac{1}{2} z^{-1} \frac{dy}{dz} + \frac{1}{4} y = 0$$

$$\frac{1}{4} \frac{d^2 y}{dz^2} + \frac{1}{4} z^{-1} \frac{dy}{dz} + \frac{1}{4} y = 0$$

Multiplying by  $4z^2$ , we get

$$z^2 \frac{d^2 y}{dz^2} + z \frac{dy}{dz} + z^2 y = 0$$

which is a Bessel's equation. It's general solution needs Bessel's functions of second kind.

General solution is

$$y = c_1 J_0(z) + c_2 Y_0(z)$$

$$y = c_1 J_0(\sqrt{x}) + c_2 Y_0(\sqrt{x})$$

Q 4. a) Consider,  $\int x^{-1} J_4(x) dx$ .

$$\text{Then, } \int x^{-1} J_4(x) dx = \int x^2 (x^{-3} J_4(x)) dx \quad \text{--- (a)}$$

$$\text{Recall: } [x^{-\nu} J_\nu(x)]' = -x^{-\nu} J_{\nu+1}(x) \quad \text{--- (b)}$$

For  $\nu=3$ , we get

$$[x^{-3} J_3(x)]' = -x^{-3} J_4(x)$$

so, (a) can be re-written as

$$\begin{aligned} \int x^{-1} J_4(x) dx &= \int x^2 (-x^{-3} J_3(x))' dx \\ &= -x^2 x^{-3} J_3(x) + \int 2x x^{-3} J_3(x) dx \\ &= -x^{-1} J_3(x) + 2 \int x^{-2} J_3(x) dx \quad \text{--- (c)} \end{aligned}$$

For  $\nu=2$  in (b), we get

$$(x^{-2} J_2(x))' = -x^{-2} J_3(x)$$

so, from (c), we get:  $-x^{-1} J_3(x) + 2 \int (-x^{-2} J_2(x))' dx$

$$\Rightarrow \int x^{-1} J_4(x) dx = -x^{-1} J_3(x) - 2 x^{-2} J_2(x) + C$$



(b)  $\text{SOP}^n$ : To show,  $J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \cos x$ .

We know:

$$J_\nu(x) = x^\nu \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{2^{2m+\nu} m! \Gamma(\nu+m+1)} \quad \text{--- (a)}$$

↓

Bessel eq<sup>n</sup> of the first kind of order  $\nu$ .

When  $\nu = \frac{1}{2}$ , we get from (a)

$$\begin{aligned} J_{\frac{1}{2}}(x) &= \sqrt{x} \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{2^{2m+\frac{1}{2}} m! \Gamma(m+\frac{3}{2})} \\ &= \sqrt{\frac{2}{x}} \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+1}}{2^{2m+1} m! \Gamma(m+\frac{3}{2})} \quad \text{--- (b)} \end{aligned}$$

To simplify the denominator,

$$\begin{aligned} \text{Consider } A &= 2^m m! = 2^m (m(m-1) \dots 3 \cdot 2 \cdot 1) \\ &= \underbrace{2 \dots 2}_{m\text{-times}} (m(m-1) \dots 3 \cdot 2 \cdot 1) \\ &= 2m \cdot (2m-2) \dots 4 \cdot 2 \end{aligned}$$

$$\text{and } B = 2^{m+1} \Gamma(m+\frac{3}{2}) = 2^{m+1} (m+\frac{1}{2}) \Gamma(m+\frac{1}{2})$$

$$\text{Again, } \Gamma(m+\frac{1}{2}) = (m-\frac{1}{2}) \Gamma(m-\frac{1}{2})$$

Repeating this down to  $\Gamma(\frac{1}{2})$ , to get

$$\Gamma(m+\frac{3}{2}) = (m+\frac{1}{2}) (m-\frac{1}{2}) \dots \frac{1}{2} \Gamma(\frac{1}{2})$$

$$\text{So, } B = 2^{m+1} \Gamma\left(m + \frac{3}{2}\right) = 2^{m+1} \left(m + \frac{1}{2}\right) \left(m - \frac{1}{2}\right) \dots \frac{1}{2} \Gamma\left(\frac{1}{2}\right) \\ = (2m+1)(2m-1) \dots 3 \cdot 1 \sqrt{\pi} \\ (\because \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi})$$

$$\text{Now, } AB = 2^{2m+1} m! \Gamma\left(m + \frac{3}{2}\right)$$

$$\Rightarrow (2m+1)(2m)(2m-1) \dots 3 \cdot 2 \cdot 1 \sqrt{\pi} = AB$$

$$\Rightarrow (2m+1)! \sqrt{\pi} = AB$$

$$\text{From (b) we get } J_{\frac{1}{2}}'(x) = \sqrt{\frac{2}{\pi x}} \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+1}}{(2m+1)!}$$

$$\Rightarrow J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$$

$$\text{Recall : } \left[ x^{\nu} J_{\nu}(x) \right]' = x^{\nu} J_{\nu-1}(x)$$

$$\text{For } \nu = \frac{1}{2}, \text{ we have } \left[ x^{\nu} J_{\nu}(x) \right]' = \left[ x^{\frac{1}{2}} J_{\frac{1}{2}}(x) \right]'$$

$$\text{So, } \left[ \sqrt{x} J_{\frac{1}{2}}(x) \right]' = \left[ \sqrt{x} \sqrt{\frac{2}{\pi x}} \sin x \right]'$$

$$\Rightarrow \sqrt{x} J_{\frac{1}{2}}(x) = \left[ \sqrt{\frac{2}{\pi}} \sin x \right]'$$

$$\Rightarrow \sqrt{x} J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi}} \cos x$$

$$\Rightarrow \boxed{J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \cos x}$$



5.  $2xy'' + (3-4x)y' + (2x-3)y = 0$  — (1)

Sol<sup>n</sup>: To determine  $\gamma_1$  and  $\gamma_2$ , we write given ODE as:

$$x^2 y'' + \left(\frac{3}{2}x - 2x^2\right)y' + \left(x^2 - \frac{3}{2}x\right)y = 0$$

So, it is of the form:

$$x^2 y'' + x b(x)y' + c(x)y = 0$$

(multiplying  
by  $x$  & dividing  
by  $2$ )

where  $b = \frac{3}{2} - 2x$ , ~~and~~  $b_0 = \frac{3}{2}$  &  $c_0 = 0$ .

Hence, we have

$$\gamma(\gamma-1) + \frac{3}{2}\gamma = \gamma\left(\gamma + \frac{1}{2}\right) = 0,$$

$$\Rightarrow \gamma_1 = 0, \gamma_2 = -\frac{1}{2}.$$

To obtain 1<sup>st</sup> sol<sup>n</sup>: we substitute  $y(x) = x^0 \sum_{m=0}^{\infty} a_m x^m$   
in (1), we get:

$$2x \sum_{m=0}^{\infty} m(m-1) a_m x^{m-2} + (3-4x) \sum_{m=0}^{\infty} m a_m x^{m-1} + (2x-3) \sum_{m=0}^{\infty} a_m x^m = 0$$

$$\Rightarrow \sum_{m=0}^{\infty} \left[ 2m(m-1) a_m x^{m-1} + 3m a_m x^{m-1} - 4m a_m x^m + 2a_m x^{m+1} - 3a_m x^m \right] = 0$$

Hence, equating the sum of the coefficients of the  
Power  $x^s$  to zero, we have

$$2(s+1) a_{s+1} + 3(s+1) a_{s+1} - 4s a_s + 2a_{s-1} - 3a_s = 0.$$

$$\Rightarrow (2s^2 + 5s + 3) a_{s-1} - (4s + 3) a_s + 2a_{s-1} = 0 \quad \text{(Combining terms \& simplifying)}$$



For  $s=0$ , as there is no coefficient  $a_{-1}$ , we obtain  $3a_1 - 3a_0 = 0$ .

Hence,  $a_1 = a_0$ . For  $s=1$ , we get

$$10a_2 - 7a_1 + 2a_0 = 0, \quad 10a_2 = 5a_0, \quad a_2 = \frac{1}{2}a_0.$$

$$\therefore (2s^2 + 5s + 3)a_{s+1} - (4s + 3)a_s + 2a_{s-1} = 0.$$

$$\Rightarrow a_{s+1} = \frac{(4s+3)a_s - 2a_{s-1}}{2s^2 + 5s + 3}$$

$$= \frac{(4s+3)a_s - 2a_{s-1}}{(s+1)(2s+3)}$$

So,

For  $s=2$ , taking  $a_0=1$ , ~~it~~ gives ~~of~~

$$\begin{aligned} a_3 &= \frac{(8+3)a_2 - 2a_1}{3 \cdot 7} = \frac{11a_2 - 2a_1}{21} \\ &= \frac{11a_2 \times \frac{1}{2} - 2 \times 1}{21} \\ &= \frac{11 - 4}{42} \\ &= \frac{7}{42} = \frac{1}{3!} \end{aligned}$$

$$\text{In general, } a_{s+1} = \frac{1}{(s+1)!}.$$

Hence the first sol<sup>n</sup>  $y_1(x) = e^x$ .

Second Sol<sup>n</sup>:

$\therefore x_1 = 0, x_2 = -\frac{1}{2}$ . We're in case 1 & have to

substitute

$y_2(x) = x^{-1/2} (A_0 + A_1 x + \dots)$  and determine the coefficients  $A_i$ 's.

using 'm' as the summation letter, as before,

$$\sum_{m=0}^{\infty} \left[ 2 \left(m - \frac{1}{2}\right) \left(m - \frac{3}{2}\right) A_m x^{m-\frac{3}{2}} + 3 \left(m - \frac{1}{2}\right) A_m x^{m-\frac{3}{2}} - 4 \left(m - \frac{1}{2}\right) A_m x^{m-\frac{1}{2}} + 2 A_m x^{m+\frac{1}{2}} - 3 A_m x^{m-\frac{1}{2}} \right] = 0$$

Hence, equating the sum of the coefficients of the power  $x^{s-\frac{1}{2}}$  to zero, we obtain

$$2 \left(s + \frac{1}{2}\right) \left(s - \frac{1}{2}\right) A_{s+1} + 3 \left(s + \frac{1}{2}\right) A_{s+1} - 4 \left(s - \frac{1}{2}\right) A_s + 2 A_{s-1} - 3 A_s = 0$$

Solving for  $A_{s+1}$  in terms of  $A_s$  &  $A_{s-1}$ , we obtain

$$A_{s+1} = \frac{1}{(s+1)(2s+1)} [(4s+1) A_s - 2 A_{s-1}]$$

Noting that there is no  $A_{-1}$  we successively ~~get~~ get,

$$A_0 = 1$$

$$A_1 = A_0 = 1 \quad (s=0), \quad A_2 = \frac{1}{2 \cdot 3} [5 A_1 - 2 A_0] = \frac{1}{2!} \quad (s=1)$$

$$\text{Similarly, } A_3 = \frac{1}{3!}, A_4 = \frac{1}{4!} \text{ etc.}$$

$$\text{So, } y_2(x) = \frac{1}{\sqrt{x}} \left[ 1 + \frac{1}{2!} x + \frac{1}{3!} x^2 + \dots \right] = \frac{1}{\sqrt{x}} e^x$$